Halting implies the entailment from the machine description to the halting sentence is valid

The Open Logic Project contributors

openlogic:turing-machines/undecidability:local-snapshot-20261013;
CC BY 4.0

Editorial note. The bridge observation that C(M,w,k) describes a halting configuration is definitional unpacking of the configuration formula's conjuncts at the halting step, retained verbatim in the counted span.

Editorial note. Both cited lemmas are corpus cards, so the composition is corpus-internal end to end.

Editorial scope. One original lemma of the unit's verification section, counted once: the halting-to-validity direction of the equivalence between the halting problem and the validity of T(M,w) implies E(M,w). The counted lemma and its author proof are retained; the reverse direction is a separate unit result outside this card.

Allowed standard prerequisites (editorial).

- The translation of a Turing machine M and input w into a first-order sentence T(M,w) describing the machine program, a halting sentence E(M,w), and the configuration formula C(M,w,n) describing the machine configuration after n steps, together with the notion of validity of a first-order sentence.
- The configuration lemma of the same unit: for each n, if M has not halted after n steps, then T(M,w) entails C(M,w,n); published as corpus card X1-logic-tm-undec-config.
- The halting-configuration lemma of the same unit: if the configuration formula describes a halting configuration (the state and scanned symbol pair has no applicable instruction), then C(M,w,k) entails E(M,w); published as corpus card X1-logic-olp-tm-halt-config-implies-halt.

Source reference. Original lemma config: sources/olp-turing-machines-undecidability/pinned-original-source.tex, source lines 1682–1682. The proof invokes the configuration lemma to obtain that T(M,w) entails the configuration description at each step before halting. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full. The lemma is proved in the same unit and is published as corpus card X1-logic-tm-undec-config.

Source reference. Original lemma halt-config-implies-halt: sources/olp-turing-machines-undecidability/pinned-original-source.tex, source lines 1648–1648. The

proof applies the halting-configuration entailment lemma to the configuration description at the halting step. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full. The lemma is proved in the same unit and is published as corpus problem 900003069 (card X1-logic-olp-tm-halt-config-implies-halt).

Lemma 0.1. *If M halts on input w, then $T(M,w) \to E(M,w)$ is valid.*

Proof. By config , we know that, for any time n, the description $C(M,w,n)$ of the configuration of M at time n is entailed by $T(M,w)$. Suppose M halts after k steps. At that point, it will be scanning square m, for some $m \in \mathbb{N}$. Then $C(M,w,k)$ describes a halting configuration of M, i.e., it contains as conjuncts both $\mathsf{Q}_q(\overline{m},\overline{k})$ and $\mathsf{S}_\sigma(\overline{m},\overline{k})$ with $\delta(q,\sigma)$ undefined. Thus, by halt-config-implies-halt , $C(M,w,k) \models E(M,w)$. But since $T(M,w) \models C(M,w,k)$, we have $T(M,w) \models E(M,w)$ and therefore $T(M,w) \to E(M,w)$ is valid. $\square$